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Find the matrix $\mathbf{B}$ for a three-node triangular element with nodal coordinates (x_1, y_1) , (x_2, y_2) , and (x_3, y_3) . Why is the matrix $\mathbf{B}$ constant along the element, and what does that imply?

Solution

First of all, a 3 node triangular element is defined by the following shape functions

$$\mathbf{N} = \begin{pmatrix} 1 - \xi - \eta \\ \xi \\ \eta \end{pmatrix} \quad (4.2.3.1)$$

For

$$\mathbf{x} = \sum_{i=1}^3 N_i(\xi, \eta) \cdot \mathbf{x}_i \quad (4.2.3.2)$$

$$\mathbf{u} = \sum_{i=1}^3 N_i(\xi, \eta) \cdot \mathbf{u}_i \quad (4.2.3.3)$$

$$\begin{bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \gamma_{xy} \end{bmatrix} = \mathbf{B} \cdot \begin{bmatrix} u_{1x} \\ u_{1y} \\ u_{2x} \\ u_{2y} \\ u_{3x} \\ u_{3y} \end{bmatrix} \quad (4.2.3.4)$$

$$\mathbf{B} = (\mathbf{B}_1 \quad \mathbf{B}_2 \quad \mathbf{B}_3) \quad (4.2.3.5)$$

$$\mathbf{B}_i = \begin{pmatrix} \frac{\partial N_i}{\partial x} & 0 \\ 0 & \frac{\partial N_i}{\partial y} \\ \frac{\partial N_i}{\partial y} & \frac{\partial N_i}{\partial x} \end{pmatrix} \quad (4.2.3.6)$$

It is known that

$$\frac{\partial N_i}{\partial x} = \frac{\partial N_i}{\partial \xi} \mathbf{J}^{-1} \quad (4.2.3.7)$$

$$x(\xi, \eta) = (1 - \xi - \eta)x_1 + \xi x_2 + \eta x_3 = x_1 + \xi(x_2 - x_1) + \eta(x_3 - x_1) \quad (4.2.3.8)$$

$$y(\xi, \eta) = (1 - \xi - \eta)y_1 + \xi y_2 + \eta y_3 = y_1 + \xi(y_2 - y_1) + \eta(y_3 - y_1) \quad (4.2.3.9)$$

Thus the Jacobian is

$$\mathbf{J} = \begin{pmatrix} x_2 - x_1 & y_2 - y_1 \\ x_3 - x_1 & y_3 - y_1 \end{pmatrix} = \begin{pmatrix} x_{21} & y_{21} \\ x_{31} & y_{31} \end{pmatrix} \quad (4.2.3.10)$$

$$J = \det(\mathbf{J}) = x_{21}y_{31} - x_{31}y_{21} = (x_2 - x_1)(y_3 - y_1) - (x_3 - x_1)(y_2 - y_1) \quad (4.2.3.11)$$

Which, as seen by the shoelace formula, $J = 2A$.

It is known that

$$\mathbf{J}^{-1} = \frac{\text{adj}(\mathbf{J})}{\det(\mathbf{J})} = \frac{1}{2A} \begin{pmatrix} y_{31} & -y_{21} \\ -x_{31} & x_{21} \end{pmatrix} \quad (4.2.3.12)$$

For the first node

$$\begin{pmatrix} \frac{\partial N_1}{\partial x} \\ \frac{\partial N_1}{\partial y} \end{pmatrix} = \mathbf{J}^{-1} \begin{pmatrix} \frac{\partial N_1}{\partial \xi} \\ \frac{\partial N_1}{\partial \eta} \end{pmatrix} = \frac{1}{2A} \begin{pmatrix} y_{31} & -y_{21} \\ -x_{31} & x_{21} \end{pmatrix} \begin{pmatrix} -1 \\ -1 \end{pmatrix} \quad (4.2.3.13)$$

$$\begin{pmatrix} \frac{\partial N_1}{\partial x} \\ \frac{\partial N_1}{\partial y} \end{pmatrix} = \frac{1}{2A} \begin{pmatrix} y_{21} - y_{31} \\ x_{31} - x_{21} \end{pmatrix} = \frac{1}{2A} \begin{pmatrix} y_{23} \\ x_{32} \end{pmatrix} \quad (4.2.3.14)$$

For the second node

$$\begin{pmatrix} \frac{\partial N_2}{\partial x} \\ \frac{\partial N_2}{\partial y} \end{pmatrix} = \mathbf{J}^{-1} \begin{pmatrix} \frac{\partial N_2}{\partial \xi} \\ \frac{\partial N_2}{\partial \eta} \end{pmatrix} = \frac{1}{2A} \begin{pmatrix} y_{31} & -y_{21} \\ -x_{31} & x_{21} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad (4.2.3.15)$$

$$\begin{pmatrix} \frac{\partial N_2}{\partial x} \\ \frac{\partial N_2}{\partial y} \end{pmatrix} = \frac{1}{2A} \begin{pmatrix} y_{31} \\ x_{13} \end{pmatrix} \quad (4.2.3.16)$$

For the third node

$$\begin{pmatrix} \frac{\partial N_3}{\partial x} \\ \frac{\partial N_3}{\partial y} \end{pmatrix} = \frac{1}{2A} \begin{pmatrix} y_{12} \\ x_{21} \end{pmatrix} \quad (4.2.3.17)$$

Thus, finally:

$$\mathbf{B} = \frac{1}{2}A \cdot \begin{pmatrix} y_{23} & 0 & y_{31} & 0 & y_{12} & 0 \\ 0 & x_{32} & 0 & x_{13} & 0 & x_{21} \\ x_{32} & y_{23} & x_{13} & y_{31} & x_{21} & y_{12} \end{pmatrix} \quad (4.2.3.18)$$

Note the notation $x_{ij} = x_i - x_j$.

The strain-displacement matrix $\mathbf{B}$ is uniform across the entire domain of the element Ω_e :

$$\mathbf{B}(x, y) = \text{const} \quad \forall (x, y) \in \Omega_e \quad (4.2.3.19)$$

This spatial independence explicitly confirms that the element experiences a homogeneous strain state, verifying its classification as a Constant Strain Triangle (CST).